

Target-Visibility Control Barrier Function

A minimal-intervention lean projection on the real camera plane, with a self-releasing descent governor

Soft-Precise-Landing

9 September 2026

Purpose. Keep the marker’s measured centre inside the *real* camera image plane, with a fixed buffer from the field-of-view edge, by making the smallest change to the desired inertial acceleration ${}^{\mathcal{I}}\mathbf{a}_d$ an outer loop produces. For the cross marker the visibility target is a *single point*, the crossed-lines intersection — no multi-corner containment, no marker-size term.

The command influences that point through two channels on different timescales, so it is conditioned in two tiers rather than one problem:

- **Attitude (lean).** A body tilt rotates the field of view, so the projected centre shifts *within one control step*, and depth-independently. This channel carries a hard barrier (§3) — the CBF.
- **Descent rate.** $|\tilde{\mathbf{r}}| \approx |\text{offset}|/Z$, which grows as the altitude Z shrinks. A change in the vertical command reaches the centre only two integrators later, so descent is treated as a *soft, predictive* influence: a governor (§5) that slows the descent when the centre is predicted to reach the buffer before the lateral loop can re-centre it.

Notation

Symbol	Meaning
$\tilde{\mathbf{r}}, \tilde{r}_k$	<i>measured</i> camera-frame marker centre, tangent units (px/ f), per axis $k \in \{1, 2\}$
$\mathbf{c}_{\text{next}}$	one-step predicted centre, (2)
$\mathcal{R}, f$	sensor resolution and focal length (px)
b	FoV-edge buffer fraction, CBF_BUFFER_FRAC = 0.15; $\varphi = \frac{\mathcal{R}}{2f}(1 - b)$
$L_\omega(\tilde{\mathbf{r}})$	rotational IBVS interaction matrix $\begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}$ at $(x, y) = \tilde{\mathbf{r}}$ — contains no depth
$M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	fixed 90° lean→rotation-axis map: a lean along $\mathbf{y}$ is a body rotation about $M\mathbf{y}$
$L_e = -(L_\omega M)$	effective Jacobian $\partial \mathbf{c}_{\text{next}} / \partial \mathbf{y}$
$P = R_z(90^\circ)R_z(-\psi_b)$	inertial x - $y \rightarrow$ image axes (ψ_b = body yaw; the 90° is the camera mount offset)
${}^{\mathcal{I}}\mathbf{a}_d$	desired inertial <i>specific thrust</i> (thrust/mass), NED. Hover = $-g\mathbf{e}_3$, so $\ {}^{\mathcal{I}}\mathbf{a}_d\ $ is the thrust magnitude per unit mass
$a_z = {}^{\mathcal{I}}a_{d,z} $	vertical component magnitude — an input , never a decision variable
$\mathbf{y} = P({}^{\mathcal{I}}\mathbf{a}_{d,xy}/a_z)$	commanded lean, image axes; $\mathbf{y}_d$ from the unconstrained ${}^{\mathcal{I}}\mathbf{a}_d^0$; $\mathbf{y}_{\text{curr}} = P(-R_{[2,3]}/R_{33})$ from the realized attitude R
$\mathbf{d}, \tau$	measured translational drift rate of the centre not due to our lean (tangent/s); look-ahead horizon $\tau = \text{CBF_DRIFT_TAU}$, default 0.15 s ($\tau = 0$ removes the term $\rightarrow$ reactive-only). Source: the front-end's de-rotated optic flow $\mathbf{h}_{xy}$ (identity map to the $\tilde{\mathbf{r}}$ frame), conditioned by condition_drift — residual gate / median-3 / low-pass / radial clamp
A_{cap}, g	vehicle maximum thrust-to-mass ratio; gravitational acceleration
$y_{\text{max}}, \mathbf{s}, \rho$	deliverability radius $\sqrt{A_{\text{cap}}^2/a_z^2 - 1}$; per-axis visibility slack $\mathbf{s} \geq \mathbf{0}$ and its penalty weight $\rho = \text{CBF_VIS_RHO} = 2000$
$g_z \in [g_{\text{min}}, 1]$	descent-governor scale (§5); $g_{\text{min}} = \text{CBF_GMIN} = 0.2$, $T_{\text{react}} = \text{CBF_TREAT} = 1.5$ s

1 Barrier and safe set

Per image axis k , define

$$\boxed{h_k(\tilde{\mathbf{r}}) = \varphi_k - |\tilde{r}_k|}, \quad \mathcal{S} = \{\tilde{\mathbf{r}} : h_k(\tilde{\mathbf{r}}) \geq 0, \ k = 1, 2\}. \quad (1)$$

$\mathcal{S}$ is the closed rectangle $[-\varphi_1, \varphi_1] \times [-\varphi_2, \varphi_2]$: the set of centre positions that lie on the sensor with at least the fractional margin b to each edge. $\tilde{\mathbf{r}}$ is the *measured* centre — nothing is de-rotated

onto a virtual frame. φ is a fixed camera constant. The buffer b is the robustness margin of the barrier: it absorbs, in one number, the intended visual margin, the one-step linearization residual of (2), and residual inner-loop attitude-tracking lag, so that a barrier value computed from the linear model is a lower bound on the true one to within the design regime of §2.

2 Feature response to the acceleration command

A camera rotation moves the centre by the rotational part of the IBVS interaction matrix, a standard *depth-independent* object evaluated at the measured point: $\dot{\tilde{\mathbf{r}}} = L_\omega \boldsymbol{\omega}_{\text{rp}} + \mathbf{d}$, with $\boldsymbol{\omega}_{\text{rp}}$ the body roll/pitch rate.

The quantity a lateral-acceleration command sets is not a rotation rate but the underactuated quadrotor’s *lean* — the ratio of lateral to vertical thrust, expressed in image axes, $\mathbf{y} = P(\mathcal{I} \mathbf{a}_{\text{d},xy}/a_z)$. Producing a steady lean along $\mathbf{y}$ is a body rotation about the perpendicular axis, $\boldsymbol{\omega}_{\text{rp}} = M\mathbf{y}$; a world-fixed point’s image moves opposite to that rotation, so

$$\mathbf{c}_{\text{next}}(\mathbf{y}) = \tilde{\mathbf{r}} + L_e(\mathbf{y} - \mathbf{y}_{\text{curr}}) + \tau \mathbf{d}, \quad L_e = -(L_\omega M). \quad (2)$$

The exact map of $\tilde{\mathbf{r}}$ under a finite lean change is the rotation homography of its ray; (2) is its first-order term about the current lean, and the sign of L_e is verified against an independent pinhole-plus-attitude camera model (`tools/validate_visibility_projection.py` check 1). $\mathbf{y}_{\text{curr}}$ is read from the realized attitude, so it needs no rotation from acceleration units. The $\tau \mathbf{d}$ term extrapolates measured *target* motion over the horizon τ (CBF_DRIFT_TAU, default 0.15s). Setting $\tau = 0$ removes it and the QP is purely reactive — sufficient for a stationary marker (whose centre does not drift on its own), incomplete for a moving one. At the default τ a stationary marker still sees a small lead ($\mathbf{d}$ carries the vehicle’s own approach flow; conditioned $|\mathbf{d}| \sim 0.1$, so $\tau|\mathbf{d}| \sim 0.02$ tangent) — verified regression-free against $\tau = 0$ on all seven rover motion profiles, though the IC2-5 stationary gate itself was run at $\tau = 0$.

The drift source $\mathbf{d}$. $\mathbf{d}$ is taken from the perception front-end’s translational optic flow $\mathbf{h}_{xy}$ — the de-rotated ($L_\omega \boldsymbol{\omega}$ already removed), sensor-calibrated, filtered relative-velocity signal the middle loop already uses — rather than a bespoke differencing of $\tilde{\mathbf{r}}$. $\mathbf{h}_{xy}$ is *already* in the image-tangent frame of $\tilde{\mathbf{r}}$ (the front-end applies the camera-mount swap upstream): the map is the identity, verified by regressing $d\tilde{\mathbf{r}}/dt$ on $\mathbf{h}_{xy}$ over a real approach (+ I cosine 0.87; every other signed permutation ≈ 0). Raw $\mathbf{h}_{xy}$ is unusable as a lead — on aggressive target trajectories the $\mathbf{h}$ least-squares solve throws single-frame spikes of $|\mathbf{h}_{xy}| \sim 4\text{--}16$ (vs a real drift ~ 0.1), and on a stationary target it carries $\sim 0.1\text{--}2$ of pure self-motion and sensor noise — so $\mathbf{d}$ is conditioned before use (`condition_drift`): (i) the solve’s own in-sample residual $\text{rel_resid} = \|A\hat{\mathbf{s}} - \mathbf{b}\|/\|\mathbf{b}\|$ gates the whole vector to $\mathbf{0}$ when it exceeds CBF_DRIFT_RESID_GATE (or is non-finite); (ii) a median-of-3 removes isolated spikes; (iii) a one-pole low-pass (CBF_DRIFT_LPF_ALPHA) smooths it; (iv) a radial clamp bounds $|\mathbf{d}| \leq \text{CBF_DRIFT_MAX}$ with direction preserved. Finally the lead itself is clamped so a linear extrapolation never predicts the centre crossing an axis origin — beyond that the first-order model does not hold and the prediction would ask for a reversed correction.

Validity envelope. L_e is evaluated at the measured $\tilde{\mathbf{r}}$ and held fixed for the solve. The model error of (2) is under 8% for a per-cycle lean *change* $\|\mathbf{y} - \mathbf{y}_{\text{curr}}\| \lesssim 15^\circ$. At the 50Hz control rate with a smooth outer loop this is the operating regime; a single cycle never demands a large lean

change. Empirically (4000 randomized geometries per bin): 0 true-sensor exits at $\leq 15^\circ/\text{cycle}$; a 0.35% rate at $\sim 24^\circ/\text{cycle}$ with worst overshoot +0.07 tangent, well inside $b\mathcal{R}/(2f)$.

3 Tier 1 — the visibility CBF

Each control step, given the outer loop’s unconstrained command $\mathcal{I}\mathbf{a}_d^0$ (hence $\mathbf{y}_d$) and the measurement $(\tilde{\mathbf{r}}, \mathbf{d}, R)$,

$$\boxed{\begin{aligned} (\mathbf{y}^*, \mathbf{s}^*) = \arg \min_{\mathbf{y} \in \mathbb{R}^2, \mathbf{s} \geq \mathbf{0}} \quad & \frac{1}{2} \|\mathbf{y} - \mathbf{y}_d\|^2 + \frac{\rho}{2} \|\mathbf{s}\|^2 \\ \text{s.t.} \quad & |\mathbf{c}_{\text{next}}(\mathbf{y})|_k \leq \varphi_k + s_k \quad (k = 1, 2), \quad \|\mathbf{y}\| \leq y_{\max} \end{aligned}} \quad (3)$$

with $y_{\max} = \sqrt{A_{\text{cap}}^2/a_z^2 - 1}$, after which $\mathcal{I}\mathbf{a}_{d,xy}^* = a_z P^\top \mathbf{y}^*$ and the vertical command $\mathcal{I}a_{d,z}$ is passed through unchanged. The visibility rows are affine in $\mathbf{y}$ (gradient the fixed row $L_{e,k}$); their intersection is a convex polyhedron $\mathcal{Y}$. The ball $\|\mathbf{y}\| \leq y_{\max}$ is *thrust deliverability* written in tangent form: since $\|\mathcal{I}\mathbf{a}_d\|^2 = a_z^2(1 + \|\mathbf{y}\|^2)$, it is exactly $\|\mathcal{I}\mathbf{a}_d\| \leq A_{\text{cap}}$, equivalently the lean cap $\arccos(a_z/A_{\text{cap}})$ (§4). Folding it in makes $\mathbf{y}^*$ actuator-feasible *by construction*. The per-axis slack $\mathbf{s} \geq \mathbf{0}$, penalised by $\rho \gg 1$, keeps visibility effectively hard whenever it is achievable within the ball and degrades it gracefully — rather than going infeasible — when the buffered FoV box and the deliverability ball are disjoint (near-saturated hover with the marker already far off-centre). The objective is strictly convex in $\mathbf{y}$ and the feasible set (polyhedron $\cap$ ball) is convex and always non-empty, so (3) has a unique solution.

This is a discrete-time control barrier function. The constraint of (3) is exactly $h_k(\mathbf{c}_{\text{next}}(\mathbf{y})) \geq 0$ for both k : it requires the *next-step* barrier value to be non-negative. This is the discrete-time CBF condition $h(\mathbf{x}_{k+1}) \geq \gamma h(\mathbf{x}_k)$ at $\gamma = 0$ (hard invariance), with a $\tau\mathbf{d}$ look-ahead so the invariance also anticipates exogenous target motion.

Proposition 1 (Forward invariance). *Let $\hat{\mathbf{c}}_k = \mathbf{c}_{\text{next}}$ evaluated with the realized attitude and drift of step k , and suppose (i) $\mathbf{s}^* = \mathbf{0}$ at every step (visibility is achievable within the deliverability ball — the normal regime), and (ii) the realized centre at step $k+1$ equals $\hat{\mathbf{c}}_k$ to within $b\mathcal{R}/(2f)$ (the model regime of §2). If $\tilde{\mathbf{r}}_0 \in \mathcal{S}$ then $\tilde{\mathbf{r}}_k \in \mathcal{S}$ for all k .*

Proof. By induction. If $\tilde{\mathbf{r}}_k \in \mathcal{S}$, then with $\mathbf{s}^* = \mathbf{0}$ the solution of (3) satisfies $|\hat{\mathbf{c}}_k|_k \leq \varphi_k$, i.e. $\hat{\mathbf{c}}_k \in \mathcal{S}$. By (ii) the realized $\tilde{\mathbf{r}}_{k+1}$ differs from $\hat{\mathbf{c}}_k$ by at most the buffer margin, so $|\tilde{\mathbf{r}}_{k+1,k}| \leq \varphi_k + b\mathcal{R}/(2f) = \mathcal{R}/(2f)$: the centre is still on the sensor. $\square$ The buffer thus converts the visual margin into a certified tolerance for the one-step linearization and attitude-tracking residual. (3) is *always* solvable (the slack relaxes every visibility row), so the CBF never stalls the controller; invariance is what the slack buys back, and it holds whenever the marker’s required lean fits inside the deliverability ball. The condition $\mathbf{s}^* = \mathbf{0}$ can be certified online — it is reported each step (`vis_slack`) — and is met on every frame of a nominal approach.

Proposition 2 (Pointwise-optimal, minimal intervention). *If $\mathbf{y}_d \in \mathcal{Y}$ then $\mathbf{y}^* = \mathbf{y}_d$; otherwise $\mathbf{y}^*$ is the Euclidean projection of $\mathbf{y}_d$ onto $\mathcal{Y}$, unique by strict convexity.*

The CBF is inert whenever the outer command is already visibility-safe, and otherwise perturbs it by the least $\|\mathbf{y} - \mathbf{y}_d\|$ that restores safety. In flight on a well-behaved approach the projection is a pass-through for essentially every frame.

Proposition 3 (No strangling of recovery). *Leaning toward the marker, or tangentially to it, does not increase $|\mathbf{c}_{next}|_k$ on the binding axis; such a command is left unchanged.*

Sketch. A command that reduces $|\tilde{r}_k|$ makes visibility row k inactive; the penalty term of (3) contributes zero gradient there, so a re-centring $\mathbf{y}_d$ (which has $|\hat{c}_k| \leq \varphi_k$ on every axis and $\|\mathbf{y}_d\| \leq y_{\max}$) is a stationary point and is returned unchanged.

Solver. Eliminating the slack in closed form, $s_k^* = \max(|\hat{c}_k| - \varphi_k, 0)$, turns (3) into the unconstrained-except-for-the-ball minimisation of $\frac{1}{2}\|\mathbf{y} - \mathbf{y}_d\|^2 + \frac{\rho}{2} \sum_k \max(|\hat{c}_k(\mathbf{y})| - \varphi_k, 0)^2$, a smooth strictly-convex function of $\mathbf{y} \in \mathbb{R}^2$. It is solved by projected Newton: the Hessian is constant on each active set (identity plus $\rho L_{e,k}^\top L_{e,k}$ over the violated rows), so a handful of steps reach the interior optimum exactly; if the ball binds — only when the FoV box and the ball are disjoint — the boundary optimum is finished with a one-dimensional search over the circle $\|\mathbf{y}\| = y_{\max}$. The whole solve is a few 2×2 operations.

4 Deliverability

Thrust deliverability is the ball $\|\mathbf{y}\| \leq y_{\max} = \sqrt{A_{\text{cap}}^2/a_z^2 - 1}$ folded directly into (3). It is the exact tangent form of $\|\mathcal{I}\mathbf{a}_d\| \leq A_{\text{cap}}$, and equivalently of the lean cap $\arccos(a_z/A_{\text{cap}})$ — the true deliverable tilt at the *actual* a_z (equal to θ_{cap} only at hover; it loosens below hover, tightens above). Because a_z is a fixed input to (3), the ball is a plain norm bound on $\mathbf{y}$ and adds no coupling: $\mathbf{y}^*$, hence $\mathcal{I}\mathbf{a}_{d,xy}^*$, is actuator-feasible with no post-hoc rescaling. When box and ball conflict, the slack (not a hard clip) absorbs the visibility deficit, and $\mathbf{y}^*$ sits on the ball at the point of least remaining $\sum_k s_k^{*2}$ — the best-effort in-frame command the actuator can deliver.

Two guards remain outside (3), both normally inert: a repeat of the lean cap and the full-vector cap $\|\mathcal{I}\mathbf{a}_d\| \leq A_{\text{cap}}$ (they cover the raw command path and the degenerate $A_{\text{cap}} \leq g$ case, in which the ball is empty and θ_{cap} governs instead); and Tier 2 (§5), which only ever *shrinks* $|\mathcal{I}a_{d,z}|$ toward hover, so it cannot push $\mathcal{I}\mathbf{a}_d$ back outside the ball.

5 Tier 2 — the descent-rate governor

A change in a_z does not move $\mathbf{c}_{next}$ this cycle, so it lies outside the CBF. It is handled as a soft governor. Let, per axis,

$$\text{rem}_k = \varphi_k - |\tilde{r}_k|, \quad \text{close}_k = \max(\text{sgn}(\tilde{r}_k) \dot{\tilde{r}}_k, 0), \quad t_{\text{edge}} = \min_k \frac{\text{rem}_k}{\text{close}_k}, \quad (4)$$

the earliest predicted time for the centre to reach the buffer at its measured outward image speed ($t_{\text{edge}} = \infty$ if it is not moving outward). Then

$$\boxed{g_z = g_{\min} + (1 - g_{\min}) \text{clip}\left(\frac{t_{\text{edge}}}{T_{\text{react}}}, 0, 1\right), \quad \mathcal{I}a_{d,z}^{**} = (\mathcal{I}a_{d,z}^* + g)g_z - g,} \quad (5)$$

applied only while a descent is commanded ($\mathcal{I}a_{d,z}^* + g > 0$) and low-passed so a_z does not step. Only the *downward* part of the vertical command is scaled; $g_{\min} > 0$ keeps a minimum descent and the map cannot command a climb.

Self-releasing. g_z is monotone increasing in t_{edge} , and t_{edge} increases as $|\tilde{\mathbf{r}}|$ decreases (larger rem, smaller close). Slowing the descent gives the lateral loop more cycles at a given altitude to reduce $|\tilde{\mathbf{r}}|$; as it does, $t_{\text{edge}} \rightarrow \infty$ and $g_z \rightarrow 1$. The trigger is an external *measured* quantity ($|\tilde{\mathbf{r}}|$ and its rate), so the governor cannot self-sustain on a quantity it generates, and it is applied after Tier 1 so it never enters the lean solve. It is not a barrier and makes no invariance claim; its role is to widen the time window in which Proposition 1’s $\mathbf{s}^* = \mathbf{0}$ condition holds near touchdown.

6 Assumptions

Assumption 1. *The camera-mount rotation P and the coupling $L_e = -(L_\omega M)$, with its sign, are calibrated (validated against an independent pinhole model).*

Assumption 2. *The conditioned command is realized: the inner attitude loop tracks the attitude built from $\mathbf{y}^*$ on a timescale fast relative to the outer loop, with residual lag inside the buffer b .*

Assumption 3. *$\mathbf{d}$ is the translation part of the observed centre velocity (from the front-end’s de-rotated optic flow $\mathbf{h}_{xy}$, identity-mapped), after **condition_drift**: residual-gated to $\mathbf{0}$ on a low-confidence solve, median- and low-pass-filtered, and radially clamped. The default $\tau = 0.15$ s enables the lead (regression-free vs $\tau = 0$ across seven rover profiles); $\tau = 0$ restores reactive-only. Whether the lead improves moving-target visibility is not yet established — it needs a rover approach dynamically stable enough to observe.*

Assumption 4. *The per-cycle lean change stays within the first-order regime of §2 ($\lesssim 15^\circ$), which the 50 Hz rate and a smooth outer loop provide.*

Assumption 5. *A_{cap} and g are fixed vehicle/physical constants, used directly.*

7 Scale-freeness and depth-freeness

Every quantity in (1), (2), (3), (4)–(5) is one of: a fixed camera intrinsic ($\mathcal{R}/(2f)$, b), an attitude (R , ψ_b), an image-space measurement ($\tilde{\mathbf{r}}$, $\dot{\tilde{\mathbf{r}}}$, $\mathbf{d}$), a ratio or pass-through of the outer loop’s own command ($\mathbf{y}$, a_z), a fixed physical constant (A_{cap} , g), or a fixed time constant (τ , T_{react}). L_ω is depth-independent by construction. t_{edge} is an image distance over an image speed — seconds, dimensionless in scale. No target range, altitude, or metric velocity appears anywhere. The formulation is manifestly scale- and depth-free.